

Data Structures and Program Design Using C - SCS 1301

Exercise

Consider the following C program (info.c):

```
#include <stdio.h>
#include <limits.h>
#include <float.h>

// Program to display system and type information
int main() {
    int    2count = 10;           /* invalid variable name */
    int    item count = 20;       /* invalid variable name */
    int    _itemCount = 30;       /* valid variable name   */

    int    age    = 23;
    double gpa    = 3.856712345;
    char   grade  = 'A';

    // --- sizeof output ---
    printf("int      : %zu byte(s)\n", sizeof(int));
    printf("double   : %zu byte(s)\n", sizeof(double));
    printf("char     : %zu byte(s)\n", sizeof(char));

    // --- limits output ---
    printf("int max: %d\n", INT_MAX);
    printf("dbl max: %e\n", DBL_MAX);

    // --- formatted output ---
    printf("%10d\n", age);
    printf("%-+10d\n", age);
    printf("%010d\n", age);
    printf("%.3f\n", gpa);
    printf("%E\n", gpa);
    printf("%c = %d\n", grade, grade);

    return 0;
}
```

Answer the following questions based on the program above:

- The program contains three variable declarations marked with comments. Identify which are **valid** and which are **invalid**, and explain why the invalid ones are not allowed.
- Fix the invalid variable declarations so the program compiles. Then run the program and write down the output produced by the three *sizeof* lines.
- What values are printed by the two *limits output* lines? Using `<limits.h>` and `<float.h>`, add two more lines to also print the minimum value of `int` and the maximum value of `float`.
- Run `gcc -E info.c`. What happened to the two comments (`//` and `/* */`) in the preprocessor output?
- Run `gcc -S info.c`. Find where `printf` is called in the `.s` file. Write down the assembly instruction used to invoke it.
- Write the exact output produced by each of the six *formatted output* lines in the program (the six `printf` calls after the *limits* section).